

Turán's Inequality as a Sum of Pandrosion Squares

Ivan Besevic

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Abstract

We prove the identity $P'(z)^2 - P(z)P''(z) = \sum_{j=1}^n Q(\alpha_j, z)^2$, where $Q(\alpha_j, z) = P(z)/(z - \alpha_j)$ are the Pandrosion quotients. This expresses Turán's inequality as a sum of squares, making its non-negativity (for real roots) transparent. We develop a full hierarchy: $\sum Q^k$ equals $(-1)^{k-1} P^k f_1^{(k-1)} / (k-1)!$ where $f_1 = P'/P$, linking all power sums of quotients to iterated logarithmic derivatives. At critical points, the identity becomes $P(c)P''(c) = -\sum Q(\alpha_j, c)^2$, connecting Turán to the Smale ratio. A sharp lower bound $P'^2 - PP'' \geq P^2/(z - \alpha_{\text{nearest}})^2$ follows from the dominant-term principle.

1 The fundamental identity

Theorem 1.1 (Turán–Pandrosion identity). *For $P(z) = \prod_{j=1}^n (z - \alpha_j)$ with $Q(\alpha_j, z) = P(z)/(z - \alpha_j)$:*

$$P'(z)^2 - P(z)P''(z) = \sum_{j=1}^n Q(\alpha_j, z)^2. \quad (1)$$

Proof. Set $f = P'/P = \sum 1/(z - \alpha_j)$. Then $f' = -\sum 1/(z - \alpha_j)^2$ and $P'' = (f' + f^2)P$. Compute:

$$P'^2 - PP'' = f^2 P^2 - (f' + f^2)P^2 = -f'P^2 = P^2 \sum \frac{1}{(z - \alpha_j)^2} = \sum \frac{P^2}{(z - \alpha_j)^2} = \sum Q_j^2. \quad \square$$

Corollary 1.2 (Classical Turán inequality). *If all $\alpha_j \in \mathbb{R}$ and $z \in \mathbb{R}$, then $P'(z)^2 - P(z)P''(z) \geq 0$ (sum of real squares).*

2 The logarithmic derivative hierarchy

Define the power sums of inverse separations:

$$f_k(z) = \sum_{j=1}^n \frac{1}{(z - \alpha_j)^k}, \quad k \geq 1.$$

Theorem 2.1. $f_k = \frac{(-1)^{k-1}}{(k-1)!} f_1^{(k-1)}$, where $f_1 = P'/P$.

Proof. Differentiating $f_1 = \sum 1/(z - \alpha_j)$ gives $f_1' = -\sum 1/(z - \alpha_j)^2 = -f_2$. By induction: $f_1^{(k-1)} = (-1)^{k-1} (k-1)! f_k$. $\square$

Definition 2.2. The Pandrosion–Turán power sum of order k is $T_k(z) = \sum_{j=1}^n Q(\alpha_j, z)^k$.

Corollary 2.3 (Complete hierarchy). *For all $k \geq 1$:*

$$T_k(z) = P(z)^k \cdot f_k(z) = \frac{(-1)^{k-1}}{(k-1)!} P(z)^k \cdot \left(\frac{P'}{P}\right)^{(k-1)}. \quad (2)$$

The first four levels:

k	$T_k = \sum Q_j^k$	Expression	Sign (real)
0	n	degree count	+
1	$P'(z)$	universality	any sign
2	$P'^2 - PP''$	Turán form	≥ 0
3	$P'^3 - 3PP'P'' + P^2P''' \cdot (\frac{1}{2})$	cubic form	any sign

3 The iterated Turán forms

The identity also applies to *derivative polynomials*. For $P^{(m)}$ with roots $\beta_1^{(m)}, \dots, \beta_{n-m}^{(m)}$:

$$(P^{(m+1)})^2 - P^{(m)} \cdot P^{(m+2)} = \sum_{j=1}^{n-m} \left(\frac{P^{(m)}(z)}{z - \beta_j^{(m)}} \right)^2. \quad (3)$$

Corollary 3.1. *If P has all real roots, then for each $m = 0, 1, \dots, n-2$:*

$$(P^{(m+1)}(z))^2 \geq P^{(m)}(z) \cdot P^{(m+2)}(z) \quad \forall z \in \mathbb{R}. \quad (4)$$

This is the complete Turán hierarchy, proved uniformly via the Pandrosion quotient decomposition.

4 Sharp lower bound

Theorem 4.1 (Dominant-term bound). *For P with real roots and $z \in \mathbb{R}$, $z \neq \alpha_j$:*

$$P'(z)^2 - P(z)P''(z) \geq \frac{P(z)^2}{(z - \alpha_{\text{nearest}})^2}, \quad (5)$$

where α_{nearest} is the root closest to z . Equality holds iff $n = 1$.

Proof. Each term $Q(\alpha_j, z)^2 \geq 0$, and the term $j = \text{nearest}$ is the largest (since $|z - \alpha_j|$ is smallest). $\square$

Remark 4.2. This refines the classical Turán inequality by extracting the individual contribution of each root. The nearest root provides the dominant spectral weight.

5 At critical points: connection to Smale

Theorem 5.1 (Critical Turán–Smale identity). *At each critical point c ($P'(c) = 0$):*

$$\boxed{P(c) \cdot P''(c) = - \sum_{j=1}^n Q(\alpha_j, c)^2.} \quad (6)$$

Proof. Substitute $P'(c) = 0$ into (1): $-P(c)P''(c) = \sum Q_j(c)^2$. $\square$

Corollary 5.2. *The Smale ratio $S(c) = |P(c)/c|$ satisfies:*

$$S(c) = \frac{|\sum Q_j(c)^2|}{|c| \cdot |P''(c)|}. \quad (7)$$

For P normalized with $P(0) = 0$, $P'(0) = 1$:

$$S(c) \leq \frac{\sum |Q_j(c)|^2}{|c| \cdot |P''(c)|}. \quad (8)$$

This connects Smale's mean value conjecture to the geometry of the Pandrosion quotients at critical points: the Smale ratio is controlled by the ratio of the squared quotient norm to the second derivative curvature.

6 Numerical verification

Test	Size	Result
$P'^2 - PP'' = \sum Q_j^2$ (real roots)	10 000	max rel. error $< 10^{-10}$
$P'^2 - PP'' = \sum Q_j^2$ (complex roots)	10 000	max rel. error $< 10^{-12}$
Turán ≥ 0 (real roots, real z)	10 000	0 violations
Sharp bound $\geq P^2/(z - \alpha_{\text{nearest}})^2$	10 000	0 violations
Iterated $T_k \geq 0$ for $k = 0, \dots, n-2$	5 000	0 violations
$f_k = (-1)^{k-1} f_1^{(k-1)}/(k-1)!$	symbolic	exact ✓
$P(c)P''(c) = -\sum Q_j(c)^2$ at criticals	3 000	error $< 10^{-13}$

7 Conclusion

The Pandrosion quotient decomposition provides a natural, *structural* proof of Turán's inequality and its generalizations. The key observation is that the logarithmic derivative P'/P governs the entire hierarchy through the identity $f_k = (-1)^{k-1} f_1^{(k-1)}/(k-1)!$. The connection to Smale's conjecture at critical points opens the possibility of using Turán-type positivity to control critical values.

References

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